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رقم الكمبيوتر * 

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الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس) 

DQBL

 **Will be reviewed**

8

1s complement is used as checksum in _____. 

☐ TCP and HTTP

- ☐ TCP but not UDP
- ☐ UDP but not TCP
- ☒ Both TCP and UDP

 **Will be reviewed**

9

FTP runs over UDP – need to ensure files are uploaded/downloaded intact; 

- ☒ True
- ☐ False

 **Will be reviewed**

10

Which of the following is present in both HTTP request line and status line? 

- ☐ URL
- ☒ HTTP version number
- ☐ Request method
- ☐ Status code

 **Will be reviewed**

11

Host A sends 5 data packets to host B using TCP protocol. Each data packet contains 10 bytes of application data. Suppose 5 data packets arrive in order and all are accepted by B . The last ACK packet sent by B has ACK number 99. What is the sequence number of the first data packet sent by A ? 

- ☐ 58
- ☐ 100
- ☐ 99
- ☒ 49

 **Will be reviewed**

12

If the fragment offset in an IP header has a value of 100, it means that _____.



- ☐ this IP datagram is 100 bytes in size
- ☐ this IP datagram is not a fragment
- ☐ this first byte of payload in the IP datagram is byte 100
- ☒ this first byte of payload in the IP datagram is byte 800

 **Will be reviewed**

13

Suppose the propagation delay between furthest nodes is d and link rate is r . What is the minimal frame size L to ensure collision will always be detected in

CSMA/CD protocol? 

- ☐ $2 * d * r$
- ☐ $2 * d + r$
- ☐ $d + r$
- ☒ $2 * (d + r)$

 **Will be reviewed**

14

Which of the following protocols run at the application layer?

- i) SMTP
- ii) UDP
- iii) POP
- iv) DNS 

- ☐ (i) and (iv) only
- ☒ (i), (iii) and (iv) only
- ☐ (i), (ii) and (iv) only
- ☐ (i) and (iii) only

 **Will be reviewed**

15

What is the 1s complement of the sum of the following 2 bytes: **11011010** and **01100101**? 

☐ 00111110☐ 11000001☐ 01000000☒ 10111111

 **Will be reviewed**

16

In a subnet, the first IP address is 172.18.176.0 and the last IP address is 172.18.183.255. What is the length of network prefix of this subnet? 

☐ 29☐ 28☐ 22☒ 21

 **Will be reviewed**

17

The access method for wireless LAN as defined by IEEE 802.11 is based on _____. 

☐ CSMA/CD☐ Polling☐ CSMA



☑ **Will be reviewed**

18

How many of the following IP addresses belong to the subnet 192.168.160.0/20?

- i) 192.168.15.1
- ii) 192.168.177.254
- iii) 192.188.168.230
- iv) 192.168.169.31



☐ 0

☐ 3

☒ 1

☐ 2

☑ **Will be reviewed**

19

The _____ layer of the Internet protocol stack is responsible for delivering data from sending process to receiving process. 

☐ application

☒ transport

☐ network

☐ link

 **Will be reviewed**

20

Which of the following is the Web's application layer protocol? 

☒ HTTP

☐ UDP

☐ TCP

☐ DHCP

 **Will be reviewed**

21

What does HTTP stand for? 

☐ Hypertext Translation Protocol

☒ Hypertext Transfer Protocol

☐ Hypertext Transition Protocol

☐ Hypertext Transmission Protocol

 **Will be reviewed**

22

The Internet protocol stack consists of _____ protocol layers. 

☐ 7☐ 6☐ 4☒ 5

 **Will be reviewed**

23

Which of the following is a VALID subnet mask? 

☐ 224.0.0.0☐ 255.255.255.128☒ 255.254.0.1☐ 254.240.0.0

 **Will be reviewed**

24

Which of the following statements regarding the Internet is TRUE? 

☐ A call setup is always performed before data transmission starts.☒ The Internet is structured as a network of networks.

- ☐ The only access network technologies allowed are Ethernet and Wi-Fi.
- ☐ A packet passes through no more than two autonomous systems to reach destination host.

 **Will be reviewed**

25

The _____ field in IP header prevents an IP datagram from continuously wandering through the Internet. 

- ☐ upper layer protocol
- ☐ header checksum
- ☒ destination IP address
- ☐ TTL

 **Will be reviewed**

26

Which of the following layers doesn't appear in the Internet Protocol Stack?



- ☒ session
- ☐ transport
- ☐ application
- ☐ network

☑ Will be reviewed

27

Two hosts A and B are 2000 km apart and are connected by a direct link of 200 kbps. Propagation speed on the link is 2.5×10^8 m/s. A is sending 10 packets to B continuously. Each packet is 100 bytes in size. Suppose the channel is error-free, how long (in seconds) does it take for B to receive all the packets? 📎

- ☐ 0.084
- ☒ 0.048 seconds
- ☐ 0.008 seconds
- ☐ 0.04 seconds

☑ Will be reviewed

28

Can HTTP response message contains an empty body? If yes, When can this happen? 📎

- ☐ Yes, if the requested object is recently modified.
- ☐ No, an HTTP response message can never have an empty body.
- ☐ Yes, if the requested object is very small in size.
- ☒ Yes, if the requested object has been moved to a new location.

☑ Will be reviewed

29

An IP datagram with destination address 192.168.1.10 reaches a router with the following 4 routing entries. Which entry will be used to forward this packet?



- ☒ 192.168.1.0/24
- ☐ 192.168.8.0/22
- ☐ 192.168.1.16/27
- ☐ 192.168.0.0/23

 **Will be reviewed**

30

Consider a datagram network using 8-bit IP addresses. Suppose a router uses longest prefix matching and has the following forwarding table:

For interface 4, give the associated range of destination IP addresses and the number of destination IP addresses in that range.



- ☐ 1000 0000 – 1001 1111, 32
- ☐ 0000 0000 – 0111 1111, 128
- ☒ 1010 0000 – 1011 1111, 32
- ☐ 1101 0000 – 1101 1111, 16

 **Will be reviewed**

31

The _____ layer is responsible for moving datagrams from the source host to the destination host. 

- ☐ application
- ☐ link
- ☐ transport
- ☒ network

 **Will be reviewed**

32

Live video streaming often runs over TCP – fast; relatively stable throughput!



- ☒ True
- ☐ False

 **Will be reviewed**

33

Consider sending a sequence of packets from a host in Assiut University to another host in Assiut National University. Packets may be of different length but all go through the same route to the destination. Which of the following end-to-end delay component is a constant (i.e. doesn't vary from packet to packet)? 

- ☒ Processing delay

- ☐ Queueing delay
- ☐ Propagation delay
- ☐ Transmission delay

✉ **Will be reviewed**

34

Knowing that you have taken Computer Networks, a friend approaches you for help with his laptop. He says that he cannot access the Web page hosted at www.example.org. Using the tools you have learned in Computer Networks, you run the following commands on his laptop to troubleshoot what could be the reason. 

- ☐ You run telnet to check if www.example.org is listening on port 80.
- ☐ You run traceroute to check if there is a route from the laptop to www.example.org.
- ☐ You run ping to check if you can establish a TCP connection to www.example.org.
- ☒ You run dig to check if his DNS server is able to resolve the IP address of host name www.example.org.

✉ **Will be reviewed**

35

Which of the following best characterizes CSMA/CD protocol? 

- ☐ Listen to the channel before accessing it and continue listening to the channel while the transmission is in progress.
- ☒ Listen to the channel before accessing it.

- ☐ Send whenever there is data ready to send.
- ☐ Send whenever there is data and then listen to the channel for collision.

📝 **Will be reviewed**

36

CSMA/CD is a (an) _____ layer protocol. 📎

- ☐ transport
- ☐ application
- ☒ link
- ☐ network

📝 **Will be reviewed**

37

For the following screenshot taken on a hand phone, what is the IP address of the hand phone in CIDR format? 📎

- ☐ 117.91.224.0/22
- ☐ 117.91.226.234/22
- ☒ 117.91.224.1/22
- ☐ 117.91.224.0/24

📝 **Will be reviewed**

38

A sender and a receiver communicate over an unreliable channel of the following characteristics: **packets may be lost or delayed for arbitrarily long time but won't be corrupted or reordered.**

We would like to design a stop-and-wait, NAK-free protocol called rdt 4.0 to ensure reliable transmission. In rdt 4.0, sender keeps a timer for a sent but unacknowledged packet. When timer expires, sender deems the packet is lost and will retransmit it.

You are to design other features of rdt 4.0 to make it a reliable protocol for the given scenario. The features you design must not conflict with the features/specifications given above.

Which of the following statements about your rdt 4.0 is TRUE?



- ☒ Receiver must maintain a timer for the ACK sent. If no packet arrives before timer expires, receiver should retransmit the ACK.
- ☐ Sender must resend data packet if a duplicate ACK is received.
- ☐ Sender won't receive duplicate ACKs at all.
- ☐ Receiver must send ACKs for duplicate packets.

Will be reviewed

39

In a host that is sending data to the network, UDP receives data from the _____ layer and sends data to the _____ layer.

- ☐ transport; network
- ☐ network; application
- ☐ application; network

☒ application; transport

 **Will be reviewed**

40

Which of the following statement regarding TCP is TRUE? 

- ☐ If a TCP segment has sequence number m , then ACK for this segment must have acknowledgement number $m + 1$.
- ☐ Host A is sending a file to Host B over a TCP connection. If the sequence number of a TCP segment is m , then the sequence number of the subsequent segments must be greater than m .
- ☐ If a TCP segment has sequence number m , then ACK for this segment must have acknowledgement number m .
- ☒ TCP provides a connection-oriented service to its applications.

 **Will be reviewed**

41

In circuit switching, circuit segment is idle if not used by call. 

- ☒ Agree
- ☐ Disagree

 **Will be reviewed**

42

A host uses a variety of protocols to discover information about the network it is connected to. Which of the following statements is FALSE? 

- ☒ To perform a DNS lookup, a host must first discover the IP address of its local DNS server using DHCP.
- ☐ To send a packet outside the host's subnet, a host must first discover the IP address of the destination host using DNS.
- ☐ To get an IP address assigned, a host must first discover the IP address of its DHCP server using DNS.
- ☐ To send a packet outside the host's subnet, the host must first discover the IP address of its first-hop router using DHCP.

 **Will be reviewed**

43

The first line in an HTTP request message is called the _____ line. The first line in an HTTP response message is called the _____ line. 

- ☐ status; status;
- ☐ request; status;
- ☐ request; header;
- ☒ request; response;

 **Will be reviewed**

44

Is it necessary for an IP datagram to carry both the source and the destination IP addresses? 

☐ No☒ Yes

 **Will be reviewed**

45

Which of the following makes up the end-to-end delay? 

☐ processor delay, processing delay, queueing delay, transmission delay☐ packing delay, propagation delay, receiving delay, transmission delay☒ processing delay, propagation delay, queueing delaying, transmission delay☐ routing delay, propagation delay, queueing delay, receiving delay

 **Will be reviewed**

46

Which of the following fields is present in both UDP header and TCP header?

☒ Checksum☐ Destination IP address☐ Acknowledgement number

☐ Sequence number

 **Will be reviewed**

47

The data value to be transmitted is **10100110** and the CRC generator is **11001**.
What is the CRC value computed? 

☐ 1001

☐ 0010

☐ 0101

☒ 1100

 **Will be reviewed**

48

Ethernet provides an unreliable service. Therefore, 

☒ CRC is not used for error checking

☐ Ethernet drops a frame that fails error checking without retransmission

☐ Ethernet sends a negative acknowledgement to the sender to indicate packet loss.

☐ Ethernet does not function correctly when bit errors in frames are detected

 **Will be reviewed**

49

Suppose there is a 10 Mbps microwave link between a geostationary satellite and its base station on Earth, which are 3.6×10^7 meters apart. The satellite takes a digital photo once in a while and then sends it to the base station. Assume a propagation speed of 2.4×10^8 meters/second. What is the propagation delay (in seconds) of the link? Suppose the satellite takes a photo every 24 seconds and let x denote the size of the photo. What is the minimum value of x (in bits) for the microwave link to be fully utilized (i.e. satellite is always busy transmitting)? 

- ☒ The propagation delay is 0.15 seconds and the minimum value of x is 2.4×10^8 .
- ☐ The propagation delay is 0.15 seconds and the minimum value of x is 0.024×10^8 .
- ☐ The propagation delay is 1.5 seconds and the minimum value of x is 0.24×10^8 .
- ☐ The propagation delay is 0.24 seconds and the minimum value of x is 3.6×10^8 .

 **Will be reviewed**

50

Which of the following statements regarding application layer protocol is FALSE? 

- ☐ One reason DNS queries use UDP instead of TCP is that the use of TCP incurs a longer delay.
- ☒ A browser can open only one TCP connection at a time.
- ☐ HTTP typically runs over TCP.
- ☐ Applications that have very strict time constraint but can tolerate packet loss should use UDP as the underlying transport protocol.

 **Will be reviewed**

51

Which of the following technique/mechanism is most essential in the design of reliable transmission protocols? 

- ☐ Addressing
- ☐ Layering
- ☐ Encapsulation
- ☒ Error detection

 **Will be reviewed**

52

Which of the following statement about 2-dimensional parity bits is FALSE? 

- ☐ It can correct any two-bit error.
- ☐ It can correct any one-bit error.
- ☐ It can detect any one-bit error.
- ☒ It can detect any two-bit error.

 **Will be reviewed**

53

Ethernet uses a _____ physical address that is imprinted on the network interface card (NIC). 

- ☐ 48-byte

- ☐ 32-bit
- ☐ 6-byte
- ☒ 64-bit

 **Will be reviewed**

54

The _____ layer is responsible for moving frames from one hop (node) to the next. 

- ☐ application
- ☐ transport
- ☐ network
- ☒ link

 **Will be reviewed**

55

Which of the following statements about TCP/UDP checksum are TRUE?

- i) It is used to detect packet lost and corruption.
- ii) It can correct any 1-bit error.
- iii) It is computed using 1's complement.
- iv) It is computed over application data only, exclusive of TCP/UDP header.
- v) It can detect any 1-bit error. 

- ☐ (iii) and (iv) only
- ☐ (i), (iv) and (v) only

☒ (iii) and (v) only

☐ (ii) and (iii) only

 **Will be reviewed**

56

A huge file is transferred over an existing TCP connection (i.e., 3-way handshake is already done). The connection is still open after transmission. The first and last TCP segments have the sequence numbers 12,345 and 2,105 respectively. Each TCP segment carries 1,024 bytes of app data.

How many TCP segments are used to transfer the file, assuming the communication channel is perfectly reliable? 

☐ 4,194,303

☒ 10

☐ 4,194,295

☐ 4,194,294

 **Will be reviewed**

57

In Non-persistent HTTP, the client closes the TCP connection after fetching one object (one pair of HTTP request and HTTP response) 

☐ False

☒ True



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